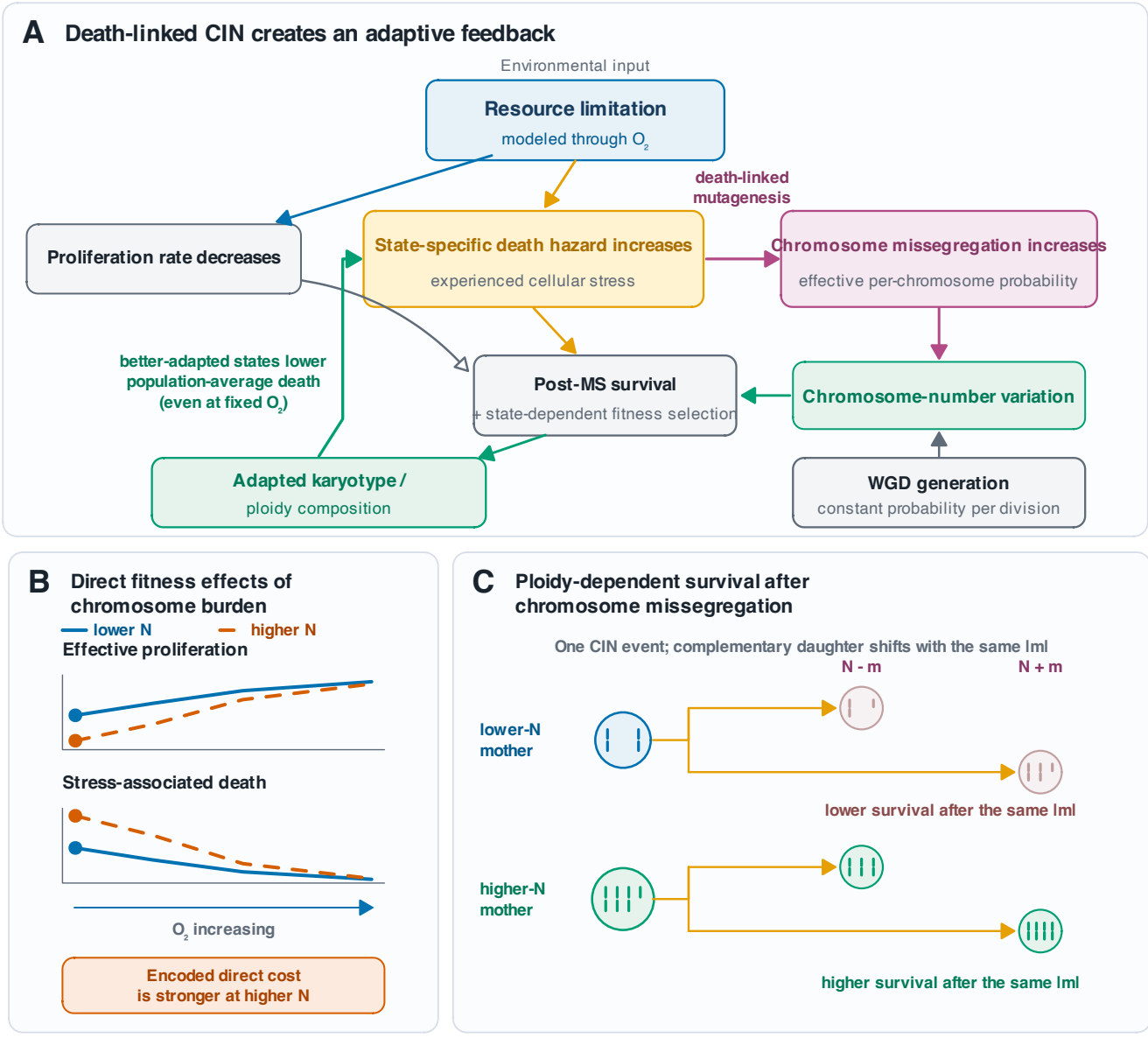
A
Death-linked CIN creates an adaptive feedback
Environmental input
Resource limitation
modeled through O_2
death-linked mutagenesis
Proliferation rate decreases
State-specific death hazard increases
experienced cellular stress
Chromosome missegregation increases
effective per-chromosome probability
better-adapted states lower population-average death (even at fixed O_2)
Post-MS survival
+ state-dependent fitness selection
Chromosome-number variation
Adapted karyotype / ploidy composition
WGD generation
constant probability per division
B
Direct fitness effects of chromosome burden
lower N
higher N
Effective proliferation
Stress-associated death
O_2 increasing
Encoded direct cost is stronger at higher N
C
Ploidy-dependent survival after chromosome missegregation
One CIN event; complementary daughter shifts with the same |m|
N - m
N + m
lower-N mother
lower survival after the same |m|
higher-N mother
higher survival after the same |m|